

Download and compile the latest linux kernel on Kylin Desktop OS v10

Target

1. Download the latest linux kernel on Kylin Desktop OS v10
 - i. linux-6.16.12.tar.xz
 - ii. <https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.16.12.tar.xz>
2. config the latest linux kernel
3. compile the latest linux kernel
4. boot ubuntu with the latest linux kernel

preparations

install related software package

```
sudo apt-get update  
sudo apt-get upgrade
```

```
sudo apt-get install build-essential
```

```
sudo apt-get install wget
```

```
sudo apt-get install pkg-config  
sudo apt-get install libgtk2.0-dev  
sudo apt-get install libcanberra-gtk-module  
sudo apt-get install glade  
sudo apt-get install libglade2-dev
```

```
sudo apt-get install flex bison
sudo apt-get install libssl-dev
sudo apt-get install libelf-dev
sudo apt-get install libncurses-dev
```

view the current linux kernel version

```
$ uname -r
5.10.0-15-generic
$
```

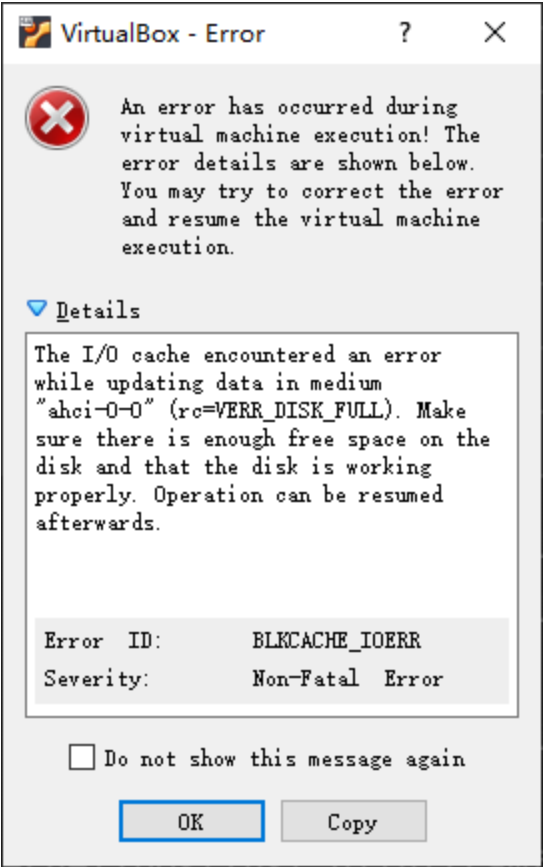


```
albert@kyosdv10: ~  
文件(F) 编辑(E) 视图(V) 搜索(S) 终端(T) 帮助(H)  
albert@kyosdv10:~$ uname -r  
5.10.0-15-generic  
albert@kyosdv10:~$ uname -a  
Linux kyosdv10 5.10.0-15-generic #1~v10pro-KYLINOS SMP Sat Oct 12 03:47:03 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux  
albert@kyosdv10:~$ cat /etc/os-release  
NAME="Kylin"  
VERSION="银河麒麟桌面操作系统V10 SP1"  
VERSION_US="Kylin Linux Desktop V10 SP1"  
ID=kylin  
ID_LIKE=debian  
PRETTY_NAME="Kylin V10 SP1"  
VERSION_ID="v10"  
HOME_URL="http://www.kylinos.cn/"  
SUPPORT_URL="http://www.kylinos.cn/support/technology.html"  
BUG_REPORT_URL="http://www.kylinos.cn/"  
PRIVACY_POLICY_URL="http://www.kylinos.cn"  
VERSION_CODENAME=kylin  
UBUNTU_CODENAME=kylin  
PROJECT_CODENAME=V10SP1  
KYLIN_RELEASE_ID="2403"  
albert@kyosdv10:~$
```

check the unused disk and ensure sufficient disk free space of "/"

```
$ df -BG
文件系统      1G的块  已用  可用  已用%  挂载点
tmpfs          1G    1G    1G    1% /run
/dev/sda3      59G   29G   28G   52% /
tmpfs          4G    0G    4G    0% /dev/shm
tmpfs          1G    1G    1G    1% /run/lock
tmpfs          4G    0G    4G    0% /run/qemu
/dev/sda2      1G    1G    1G    2% /boot/efi
tmpfs          1G    1G    1G    1% /run/user/1000
$
```

很有可能没有磁盘空间啊：建议预留40GB空闲空间
(以下是VirtualBox虚拟机的提示，VMware类似)



How to do

Download and compile the latest linux kernel

the latest linux kernel

Linux Stable Kernel: 6.16.12 (2025.10.22)

1. Download the latest linux kernel from www.kernel.org

- [linux kernel 6.16.12](#)

```
cd ~
```

```
pwd
```

```
wget -c https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.16.12.tar.xz
```

```
tar xvjf linux-6.16.12.tar.xz
```

```
sudo ln -s `pwd`/linux-6.16.12 /usr/src/linux
```

2. config the latest linux kernel

```
cd /usr/src/linux
```

```
cp /boot/config-`uname -r` .config
```

```
make oldconfig
```

此步骤需要一定的时间进行选择，一般选择N或n即可。

or: some wrong for the next command in linux-6.16.12

```
yes " " | make oldconfig
```

```
make gconfig
```

3. customized config

3.1. add local version

add your ID as local version

务必在新内核中嵌入你的姓名与学号(英文字符)!!!

Linux/x86 5.14.14 Kernel Configuration

File Options Help

Back

Load

Save

Single

Split

Full

Collapse

Expand

Options	Name	N	M	Y	Value
General setup					
☐ Compile also drivers which will not load	COMPILE_TEST	N			N
Local version - append to kernel release	LOCALVERSION				学号
☒ Automatically append version information to the version string	LOCALVERSION_AUTO			Y	Y
Build ID Salt	BUILD_SALT				
☐ Kernel compression mode					Gzip
Default init path	DEFAULT_INIT				
Default hostname	DEFAULT_HOSTNAME				(none)
☒ Support for paging of anonymous memory (swap)	SWAP			Y	Y
☒ System V IPC	SYSVIPC			Y	Y
☒ POSIX Message Queues	POSIX_MQUEUE			Y	Y
☐ General notification queue	WATCH_QUEUE	N			N
☒ Enable process_vm_readv/writev syscalls	CROSS_MEMORY_ATTACH			Y	Y
☒ uselib syscall	USELIB			Y	Y
☒ Auditing support	AUDIT			Y	Y
IRQ subsystem					
Timers subsystem					
BPF subsystem					
Documentation					Value of Kernel Documentation (Backlog)

Automatically append version information to the version string

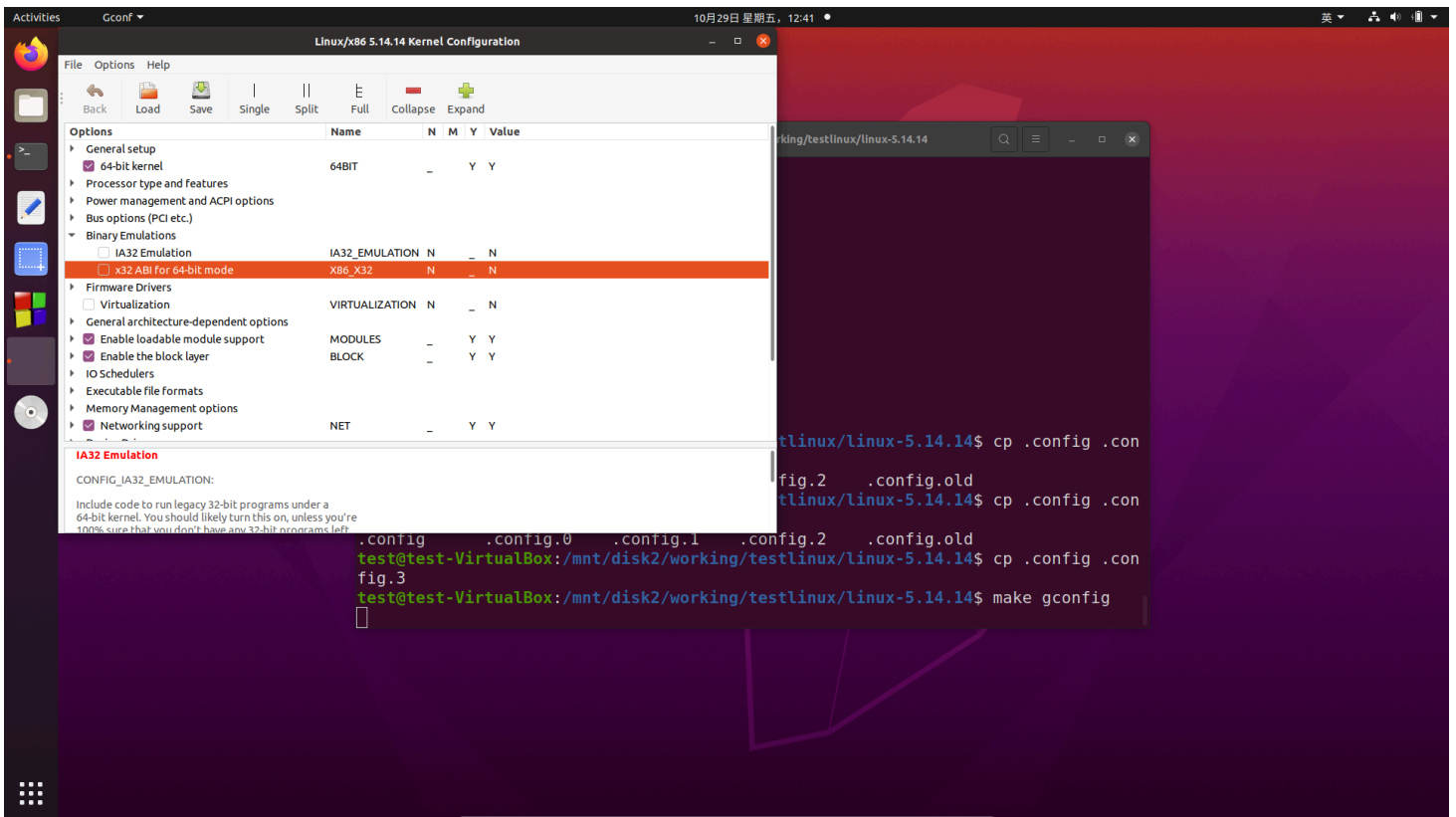
CONFIG_LOCALVERSION_AUTO:

This will try to automatically determine if the current tree is a release tree by looking for git tags that belong to the current top of tree revision.

3.2. some errors

3.2.1 arch/x86/Makefile:148: CONFIG_X86_X32 enabled but no binutils support

solution:disable the Binary Emulations



3.2.2 No rule to make target 'debian/canonical-certs.pem', needed by 'certs/x509_certificate_list

- solution: set the following keys null
 - CONFIG_SYSTEM_TRUSTED_KEYS=""
 - CONFIG_SYSTEM_REVOCATION_KEYS=""

in cryptoXXX item of config

```
# Certificates for signature checking
#
CONFIG_MODULE_SIG_KEY="certs/signing_key.pem"
CONFIG_MODULE_SIG_KEY_TYPE_RSA=y
# CONFIG_MODULE_SIG_KEY_TYPE_ECDSA is not set
CONFIG_SYSTEM_TRUSTED_KEYRING=y

#CONFIG_SYSTEM_TRUSTED_KEYS="debian/canonical-certs.pem"
CONFIG_SYSTEM_TRUSTED_KEYS=""

CONFIG_SYSTEM_EXTRA_CERTIFICATE=y
CONFIG_SYSTEM_EXTRA_CERTIFICATE_SIZE=4096
CONFIG_SECONDARY_TRUSTED_KEYRING=y
CONFIG_SYSTEM_BLACKLIST_KEYRING=y
CONFIG_SYSTEM_BLACKLIST_HASH_LIST=""
CONFIG_SYSTEM_REVOCATION_LIST=y

#CONFIG_SYSTEM_REVOCATION_KEYS="debian/canonical-revoked-certs.pem"
CONFIG_SYSTEM_REVOCATION_KEYS=""
# CONFIG_SYSTEM_BLACKLIST_AUTH_UPDATE is not set
# end of Certificates for signature checking
```

(optional)4. clean the previous compilings

If you want to recompile , please execute the following command to clean the previous work:

```
make clean
```

5. compile the latest linux kernel (long wait)

- compile the latest linux kernel

```
make -j4
```

此步骤需要执行很长时间，也有可能会出错。为了确认确实正常编译成功了，可再执行一遍make，这次会很快完成。

```

$ make -j4
...
CC      arch/x86/boot/video-vesa.o
CC      arch/x86/boot/video-bios.o
CC      arch/x86/boot/compressed/sev.o
CC      arch/x86/boot/compressed/sev-handle-vc.o
CPUSTR  arch/x86/boot/cpustr.h
CC      arch/x86/boot/compressed/acpi.o
CC      arch/x86/boot/cpu.o
CC      arch/x86/boot/compressed/mem.o
CC      arch/x86/boot/compressed/efi.o
CC      arch/x86/boot/compressed/misc.o
LZ4      arch/x86/boot/compressed/vmlinux.bin.lz4
MKPIGGY arch/x86/boot/compressed/piggy.S
AS      arch/x86/boot/compressed/piggy.o
LD      arch/x86/boot/compressed/vmlinux
ZOFFSET arch/x86/boot/zoffset.h
OBJCOPY arch/x86/boot/vmlinux.bin
AS      arch/x86/boot/header.o
LD      arch/x86/boot/setup.elf
OBJCOPY arch/x86/boot/setup.bin
BUILD  arch/x86/boot/bzImage
Kernel: arch/x86/boot/bzImage is ready  (#1)
$

```

- 再一次make,验证确实编译成功:

```

$ make
CALL      scripts/checksyscalls.sh
DESCEND  objtool
INSTALL  libsubcmd_headers
CAT      lib/test_fortify/test_fortify.log
Kernel: arch/x86/boot/bzImage is ready  (#1)
$

```

- install the latest linux kernel

```

sudo make modules_install
sudo make install

```

Successfully compiled:


```
/usr/src/linux$ sudo make install
INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/apt-auto-removal 6.16.12AL2025 /boot/vmlinuz-6.16.12AL2025
run-parts: executing /etc/kernel/postinst.d/dkms 6.16.12AL2025 /boot/vmlinuz-6.16.12AL2025
* dkms: running auto installation service for kernel 6.16.12AL2025
Kernel preparation unnecessary for this kernel. Skipping...
```

Building module:

cleaning build area...

```
make -j2 KERNELRELEASE=6.16.12AL2025 -C /lib/modules/6.16.12AL2025/build M=/var/lib/dkms/evdi/1.14.0
Error! Bad return status for module build on kernel: 6.16.12AL2025 (x86_64)
Consult /var/lib/dkms/evdi/1.14.0/build/make.log for more information.
```

Kernel preparation unnecessary for this kernel. Skipping...

Building module:

cleaning build area...

```
make -j2 KERNELRELEASE=6.16.12AL2025 -C /lib/modules/6.16.12AL2025/build M=/var/lib/dkms/optilauncher/1.0.1.0
Error! Bad return status for module build on kernel: 6.16.12AL2025 (x86_64)
Consult /var/lib/dkms/optilauncher/1.0.1.0/build/make.log for more information.
```

Kernel preparation unnecessary for this kernel. Skipping...

Running the pre_build script:

Building module:

cleaning build area...

```
make -j2 KERNELRELEASE=6.16.12AL2025 -C /lib/modules/6.16.12AL2025/build M=/var/lib/dkms/xtables-addons/3.9
Error! Build of xt_comm.ko failed for: 6.16.12AL2025 (x86_64)
Consult the make.log in the build directory
```

/var/lib/dkms/xtables-addons/3.9/build/ for more information.[OK]

```
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 6.16.12AL2025 /boot/vmlinuz-6.16.12AL2025
update-initramfs: Generating /boot/initrd.img-6.16.12AL2025
```

I: The initramfs will attempt to resume from /dev/sda7

I: (UUID=b50c71b8-0b01-4e02-986c-5a932aaa4687)

I: Set the RESUME variable to override this.

```
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 6.16.12AL2025 /boot/vmlinuz-6.16.12AL2025
Sourcing file `/etc/default/grub'
```

Sourcing file `/etc/default/grub.d/init-select.cfg'

正在生成 grub 配置文件 ...

找到主题: /usr/share/grub/themes/UKUI/theme.txt

找到 Linux 镜像: /boot/vmlinuz-6.16.12AL2025

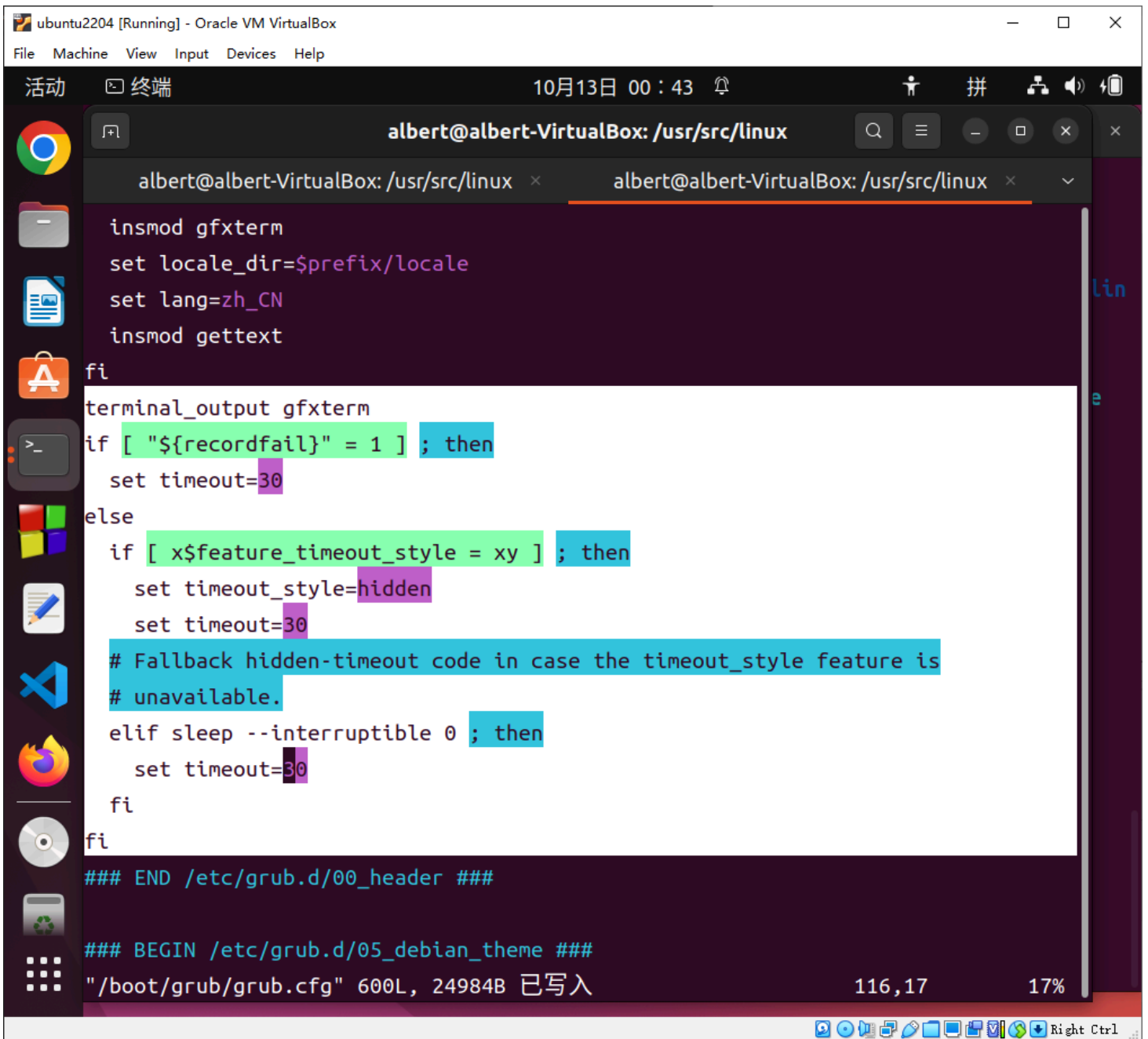
找到 initrd 镜像: /boot/initrd.img-6.16.12AL2025

```
找到 Linux 镜像: /boot/vmlinuz-5.10.0-15-generic
找到 initrd 镜像: /boot/initrd.img-5.10.0-15-generic
找到 Linux 镜像: /boot/vmlinuz-5.10.0-9-generic
找到 initrd 镜像: /boot/initrd.img-5.10.0-9-generic
找到 initrd 镜像: /boot/initrd.img-6.16.12AL2025
完成
/usr/src/linux$
```

6.(optional可选) modify the grub.cfg

```
sudo vi /boot/grub/grub.cfg
```

```
set timeout=30
```



7. boot ubuntu with the latest linux kernel

sudo reboot

uname -a

6.16.12AL2025

albert@kyosdv10: ~

— □ ×

文件(F) 编辑(E) 视图(V) 搜索(S) 终端(T) 帮助(H)

```
albert@kyosdv10:~$ uname -r
```

```
6.16.12AL2025
```

```
albert@kyosdv10:~$ uname -a
```

```
Linux kyosdv10 6.16.12AL2025 #1 SMP PREEMPT_DYNAMIC Wed Oct 22 20:23:02 CST 202
```

```
5 x86_64 x86_64 x86_64 GNU/Linux
```

```
albert@kyosdv10:~$
```

End.